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FIFTH INTERIM PROGRESS REPORT
ON THE
PHYSICAL REALIZATION OF AN
ELECTRONIC COMPUTING INSTRUMENT

12

BY

Julian H. Bigelow
Herman H. Goldstine
Richard W. Melville
Peter Panagos
James H. Pomerene
Jack Rosenberg
Morris Rubinoﬀ
Willis H.

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APPENDIX

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P R E F A C E

This report constitutes the fifth in our series of interim progress reports and covers the period from July 1, 1948, to December 31, 1948. It is prepared in accordance with the terms of Contract W-36-034-ORD-7481 between the Government of the United States and The Institute for Advanced Study.

In our four previous reports we have attempted to describe in considerable detail the underlying principles and criteria upon which the various organs would be designed. Work has now progressed to the point where much of our preliminary studies has been translated into engineering realities. Therefore our present report reflects this change in emphasis. The design considerations which influenced the choice of the circuits used in, for example, the adder, are those described in the previous reports. Therefore more and more our newer reports will describe the exact workings of our circuits and their final performance characteristics rather than the basic motivations.

All the engineering work here is under the direct supervision of Mr. Julian H. Bigelow, our chief engineer. His engineering staff consists of four electrical engineers, Messrs. J. H. Pomerene, J. Rosenberg, M. Rubinoff, and W. H. Ware, and of two mechanical engineers, Messrs. R. W. Melville and P. Panagos.

Herman H. Goldstine

1 January 1949

CHAPTER II

THE ARITHMETIC ORGAN

2.1 Introduction. In our fourth interim report there is a detailed discussion of the shifting registers, which were given their life and performance tests during the present time period. For details of the design and construction of these registers the reader should consult the fourth report, Chapter III. We give at this time the performance characteristics.

Since the writing of the last report the circuits and design characteristics for the adder were decided upon and the units themselves built and installed. Also both static and dynamic tests were made on the unit, which we give below. These tests are not complete since the gate circuits for controlling the flow of information between the registers and the adder were not complete as of January 1.

The detailed test results are given below, but it is perhaps appropriate to remark that they are entirely satisfactory and indicate that the major portion of the arithmetic organ is completed.

The tests and work described below have been and are continuing to be the joint responsibility of Messrs. Rosenberg and Ware.

2.2 The Adder. In Figure 2.1 below we give a circuit of the adder showing two stages. Before describing in detail the operation of the circuit we first discuss in more general terms its mode of operation. It is also well to once more emphasize at this point that the same principles of circuit design, which motivated the design of the shifting registers, have been embodied in the adder.

Since the arithmetic organ is to be a 40-stage "parallel type" one, the adder contains 40 essentially identical stages each of which has three inputs and two outputs. In our so-called "one-address code" we pre-suppose that in an addition the augend is already in the accumulator register and the order for an addition consists of transferring the addend into the memory register and forming the sum. The digits of the augend are indicated in Figure 2.1 as the resident digits and those of the addend as the incident digits. The latter are transferred into the adder via the so-called complement gates, which allow a number or its complement to be sent into the adder. The result of the addition, the sum, is transferred out of the adder via the so-called digit resolver gates. At the time of writing neither the complement nor digit-resolver gates have been completed in their final form.

Each stage of the adder has three inputs, one for each of the resident digits, the incident digits, and for the carry digit from the preceding stage. There are two outputs, one for the carry digit and one for the sum digit. The adder itself is of the so-called "Kirchoff" type as distinguished from the types of adders using only gate tubes. The full implications of this will be better understood as we explain in detail the functioning of the adder.

Let us consider the stage of the adder shown in the left half of Figure 2.1. It consists of the three 6J6's, numbered 1, 3, 4, the left half of the 2C51, numbered 2, and the left half of the 2C51, numbered 5. We shall call the left half of tube n n.½ and the right half n.r.

In principle it operates as follows: We bring to a summing resistor the carry digit from the previous stage as a unit of voltage, and the resident and incident digits as units of current. The sum of these three quantities 0, 1, 2, or 3 voltage units is now applied as a grid bias to a gate which is so arranged that it discriminates between the cases (0, 1) and (2, 3) giving out a voltage unit only in the latter case. This is, of course, the carry digit. The sum of the three input quantities is also fed into the so-called digit resolver which is a set of gates that decide whether the sum is either one of (1, 3) or (0, 2). In the former case, the unit digit of the answer is a 1 and in the latter it is a 0. This result is then fed into one half of the augend register displacing what was there. The digit resolver circuit will be discussed in our next report and is not shown in Figure 2.1.

The cathode follower 2.7 has as its cathode load the 10.5 k summing resistor and the plates of 3.r and 5.7 in parallel. If the resident or incident digit is a 1, then and only then 3.r or 5.7 is conducting and current flows through the 10.5k resistor. The unit of current corresponding to either tube can easily be seen to be 5 ma. Thus either 0, 5, or 10 ma will flow through tube 2.7. In the event that the carry digit is a 1, then the grid of 2.7 is dropped by 50 volts. But since a constant amount of current, 0, 5, or 10 ma, is always flowing through a fixed resistor of 10.5k, the cathode point A and the summing point S will then always move together in voltage level; and in the event that the carry digit is 1 both will drop 50 volts. This drop is precisely determined at its bottom level since the tube 1.r is so arranged as a diode that the cathode of 1.r, 1.7 cannot go below the standard level marked E_{01} . It is also fixed at its upper level

since the grid of 1.7 cannot rise above the standard voltage level to which it is tied.

The use of the two cathode followers, 1 and 2.7, in series is important so that these voltage levels can be accurately maintained. If the carry signal were applied directly to the grid of 2.7, then grid current would, in certain cases, flow and thus the upper voltage level would be destroyed.

We now see that the summing point S can have any one of four different voltage levels, differing from one another by 50 volt steps. The level of S is applied through an isolating resistor of 22k to the grid of 4.7. The grid of 4.r is so biased by the standardized voltage level E01/10 that 4.r is normally cut off and only becomes conducting if and only if at least two of the three inputs, the resident digit, the incident digit, and the carry digit, from the preceding stage, are 1. The plate of 4.r which controls whether or not a carry is to be propagated is then tied directly to the grid of the "In" carry cathode follower of the next stage.

The summing point is also connected through a 22k resistor to a tube in the digit resolver, which is not shown in the diagram, where it is decided whether there are an odd or even number of 1's being summed.

To speed up the propagation of a carry a 27 uuf condensor has been used to by-pass the 22k resistor. This allows the carry propagation to get started before a steady state obtains at the point S.

The 47k resistor in the cathode of tube 2.7 is placed there largely to prevent the cathode from attempting to come up toward the plate.

Finally the gate 4 is so biased that its gating action takes place at the midpoint of the voltage zone of 200 volts, determined by the extreme cases of three zeros and three ones. It should be remarked that small changes in this bias level will affect the rate at which a carry propagates or "collapses" -- this term is used to describe the process of clearing a carry chain out of the adder, i.e., the process of closing all relevant gates 4. Clearly the nearer the bias level of gate 4 is to the boundary of the 50 volt zone determined by three zeros and two zeros plus a one, the more quickly will a carry propagate and the longer will a collapse take.

The operational tests of the adder will be performed during the next period, i.e. after January 1, 1949, but it is of interest to state that the first of these tests showed a carry propagation time of about $6 \mu s$ and a carry collapse time of about $5 \mu s$. Furthermore the time to set the gates for an incident or resident digit is about $1 \mu s$. A life test was also performed in which the resident number was fixed at forty ones and the incident number consisted of all zeros except for the last digit which was a one introduced by a pulser. The addition, of course, propagated a carry down all stages and the carry output of the 40th stage was used to trigger the pulser. The test was run for at least eight hours at a repetition rate of about 120 kilocycles per second. No errors occurred during the tests. In the next report we shall describe the further tests of this unit in more detail.

2.3 The Registers. The shifting registers are quite completely described in our fourth interim report. Here we merely give the main results of the tests performed upon them. In order to give an adequate life test to these three registers they were interconnected in two different arrangements to form closed loops of 120 binary digits and shifted a place at a time around the loop.

In the arithmetic organ the registers appear in a vertical plane, one above the other. Let the register from top to bottom be enumerated as 1, 2, 3 as in Figure 2.2.a and Figure 2.2.b.

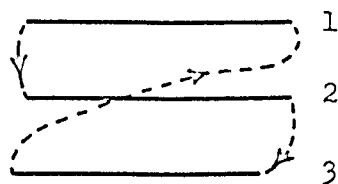


Figure 2.2.a

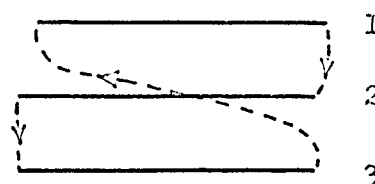


Figure 2.2.b

The arrangement in Figure 2.2.a was used for shifting the pattern to the left and in Figure 2.2.b for shifting to the right.

As was stated in the fourth report a shift consists of two distinct operations: gating the information in one half of a register into the other half and then gating it back again shifted by one place right or left. Each of these operations can safely be performed in $1.5 \mu\text{s}$ and one can succeed the other with no spacing; thus a complete shift by one place can be safely performed in $3 \mu\text{s}$.

At a rate of 3 μ s per shift, about 3.3×10^5 shifts are performed per second. At this rate the unit was operated for about 75 hours on left shifts and about 25 hours for right shifts. Thus about 10^{11} shifts in all.

A very extensive set of vibration tests were also made on the registers to find and correct bad solder joints and tubes with defective structures. A number of bad joints were detected, repaired and then no further defects in mechanical structure were found. Also three tubes with defective structures were found and replaced together with two tubes with heaters which burned out. No other tube failures occurred.